

SNAPDRAGON® AI LAB — BUILD & PRESENT CHALLENGE

Drishti

A fully offline visual assistant for blind and low-vision users — every model running on the Snapdragon Hexagon NPU.

No internet. No cloud inference. No recurring cost.

Rachit · Solo submission · github.com/Rachit0910

A large population, served by the wrong tools.

6.11%

pooled prevalence of blindness
among Indian adults aged 50 and
over

1

Dedicated hardware

Smart canes and handheld OCR readers are expensive, single-purpose, and usually imported.

2

Cloud phone apps

Upload a photo, wait 2–3 seconds, hear a sentence. Fine for a tin in a kitchen. Useless on a street.

3

The gap

Nothing offers continuous, real-time awareness that keeps working with no signal.

Prevalence: systematic review and meta-analysis, National Medical Journal of India (n > 211,000).

Drishti is the third option: continuous, real-time awareness that keeps working with no signal.

The whole design is one number.

At 30 fps a frame must be handled in 33.3 ms. Detection and depth together cost 1.29 ms on the Hexagon NPU.

1.29 ms

detection + depth,
every frame

3.9%

of the 30 fps
frame budget

~90%

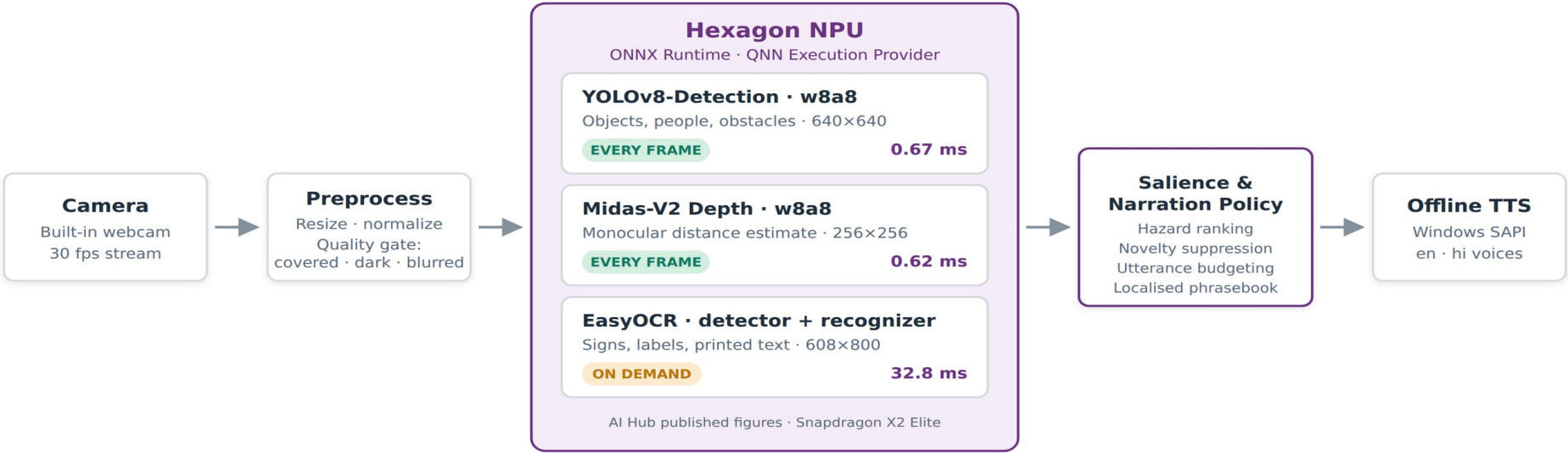
headroom left for
everything else

That headroom is the product. It is what lets a duty-cycled OCR pass, depth calibration, a salience engine and speech synthesis all share one frame without ever dropping it — on battery, for a full day.

Three AI Hub models, one always-on pipeline.

Drishti — On-Device Perception Pipeline

Every inference stage executes on the Snapdragon Hexagon NPU. No network call anywhere in the loop.



Execution fallback
QNN EP (Hexagon NPU) → DirectML (Adreno GPU) → CPU EP · identical model artifacts, graceful degradation, no rebuild

Frame budget at 30 fps = 33.3 ms
Tier A (detection + depth) = 1.29 ms → 3.9% of budget · Tier B (OCR) amortized at 2 Hz ≈ 2.2 ms/frame · steady state ≈ 10% of budget

Latencies are Qualcomm AI Hub published model-inference figures on Snapdragon X2 Elite; end-to-end pipeline latency additionally includes capture, pre/post-processing and NMS.

Three off-the-shelf models is the easy part.



Relative depth is not distance

Midas-V2 predicts relative inverse depth. Announcing it as metres would give confident, wrong distances. We fit the map to metric units every frame, using detected objects of known size as anchors — a person is a reliable 1.68 m ruler.



Speech is a serial channel

Pipe every detection into a TTS engine and you get “chair chair person chair”. The policy scores hazard $\times$ proximity $\times$ centrality $\times$ confidence, lets imminent hazards interrupt, and stays silent when its budget is spent.



Silence is ambiguous

An empty corridor and a covered lens sound identical. A blind user cannot tell them apart — so degraded input is announced, not swallowed.

All three are implemented and unit-tested today — they run on any machine, with no camera and no NPU.

A covered lens and an empty room sound the same.

Drishti's correct behaviour in an empty corridor is silence. Its behaviour with a thumb over the lens is also silence. A sighted user glances at the camera. The only user this product has cannot.

Dark and featureless

"Camera is covered."

→ move your hand

Dark but textured

"Too dark to see."

→ turn on a light

No sharp edges

"Camera image is blurred."

→ hold steadier

Recovered

"Camera clear."

→ silence is meaningful again

Three conditions, not one, because the remedy differs. Debounced so a single dropped frame never cries wolf.

This is not a plan. It runs today.

37

unit tests passing, on any laptop — no camera, no NPU, no account



Narration policy engine

17 tests



Frame quality gate

11 tests



Metric depth calibration

9 tests



Localisation — English + Hindi

tested

Three defects the tests caught — and changed the design

- Depth jitter across a band boundary re-announced the same obstacle several times a second.
- One frame mistaking a poster for a person shifted every announced distance by 73%.
- Darkness and occlusion were being conflated — they need different tests and different advice.

Clone the repo and run all 37 in under a minute. Everything needing Snapdragon hardware is labelled as such.

Built for the user who cannot see the screen.



Audio-first

No account, no login, no setup wizard. A flow that must be navigated visually is one this user cannot complete alone.



Speaks their language

English and Hindi ship today. No user-facing string lives in the logic — adding a language is a data change plus a system voice.



Shares the channel

Blind Windows users already run NVDA or JAWS. Short utterances, salience gating, instant yield on Escape.



Institutional scale

One signed installer across Snapdragon HP models — practical for a school for the blind or an NGO.

Every one of these follows from a single fact: the user cannot see the screen.

From published figures to measured ones.

1

Export, quantize and profile all three models on real Snapdragon silicon through AI Hub device provisioning — no hardware needed to get true numbers.

2

Validate the full loop on a Snapdragon HP PC; finish the OCR decode; confirm the context cache removes the cold start.

3

End-to-end benchmarks with a CPU baseline; sustained thermal and power run for the all-day battery claim.

4

User testing with blind and low-vision participants; native review of the Hindi phrasing; NVDA coexistence check.

Every published figure in this deck gets replaced by a measured one.